

MooMe System – Cost Estimation (12 Cows)

No	Component	Description	Quantity	Unit Price (RWF)	Total (RWF)
1	Raspberry Pi 4/5	Main controller	1	120,000	120,000
2	MicroSD Card (64GB)	Storage	1	15,000	15,000
3	Power Supply	Adapter	1	10,000	10,000
4	Non-contact Temp Sensor (IR)	Per cow	12	20,000	240,000
5	DHT22 Sensor	Environment monitoring	2	8,000	16,000
6	Air Quality Sensor (MQ-135)	Gas detection	1	15,000	15,000
7	Oxygen Sensor	Air monitoring	1	25,000	25,000
8	RFID Reader Module	Identification	1	18,000	18,000
9	RFID Tags	Per cow ID	12	2,000	24,000
10	Water Flow Sensor	Per cow intake	12	10,000	120,000
11	Load Cell + HX711	Feed measurement	12	15,000	180,000
12	Smart Milk Meter	Milk measurement	12	60,000	720,000
13	DC Water Pump	Per cow	12	20,000	240,000
14	Fans (Cooling)	Shared system	4	25,000	100,000
15	Relay Modules	Control system	5	8,000	40,000
16	Motor (Feed Dispenser)	Per cow	12	18,000	216,000
17	Pipes & Spray Nozzles	Cooling system	1 set	60,000	60,000
18	GSM Module (SIM800L)	SMS alerts	1	15,000	15,000
19	Wi-Fi Router	Connectivity	1	35,000	35,000
20	Solar Panel (200W)	Backup power	1	150,000	150,000
21	Battery (100Ah)	Energy storage	1	180,000	180,000
22	Charge Controller	Solar system	1	40,000	40,000
23	Inverter	Power conversion	1	80,000	80,000
24	Wires & Installation	Setup	1	80,000	80,000

The total cost for all components (12 cow's system) is:

₹ Total = 2,699,000 RWF

Its estimation cost may increase or decrease